AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Summer 2024-2025

Section: [*Y*]

Project Report On

*Project Name [**Super Shop Management System]***

Supervised By

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Submitted By:

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Obtained Marks for CO2 and CO3 (Description given in the following page)

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)

Evaluation Criteria (CO2)	**Total =**	**Evaluation Criteria (CO3)**	**Total =**
Requirement fulfillment		Organization of the application	
Validation		Representation and Integration of Database	
Verification		Graphical User Interface	

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenariobased project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenariobased project development and functional requirement identification. The project depicted lacks coherence or relevance to reallife scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenariobased project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenariobased project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenariobased project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive

	of verification processes, and data flow is not considered.	inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	ensuring data integrity and appropriate data flow within the system.	and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a userfriendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

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Chapter 01

INTRODUCTION:

The Super Shop Management System is a desktop-based application developed to manage the daily operations of a super shop efficiently. This system helps in handling important activities such as product management, customer information, and sales transactions.

In a traditional manual system, maintaining records on paper is time-consuming and often leads to errors, data loss, and difficulty in tracking information. It also becomes hard to manage product stock, calculate sales, and keep accurate customer records. To solve these problems, this system provides a computerized solution where all data is stored in a structured database.

The system allows the admin to control and manage all operations, including adding, updating, and deleting product information. Employees can easily process sales and record transactions. Customer details can also be stored for future reference.

The main objective of this project is to make the management process faster, more accurate, and well-organized. It also reduces data redundancy and improves overall efficiency. This system is user-friendly and suitable for real-life shop management.

Chapter 02

User Story :

Admin :

* As an Admin, I want to log into the system so that I can access and control all features. * As an Admin, I want to add, update, and delete product information so that product data stays accurate.
* As an Admin, I want to manage product stock and expiry date so that I can avoid shortages or expired items.
*As an Admin, I want to manage employee accounts so that I can control who can access the system.

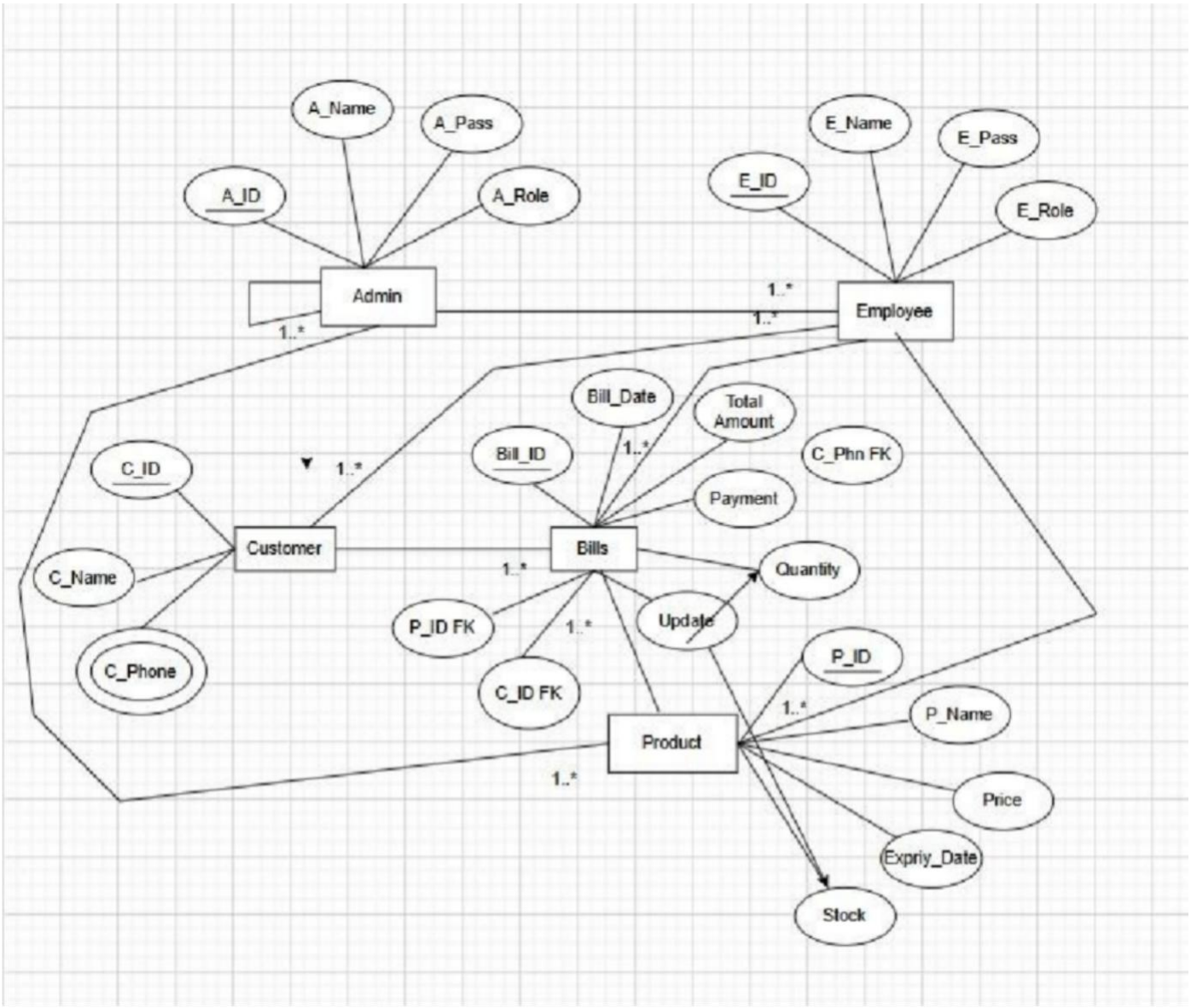
Employee :

* As an Employee, I want to log into the system so that I can perform my work securely. * As an Employee, I want to select products and enter quantity during sales so that I can complete customer purchases.
*As an Employee, I want to generate sales records so that all transactions are stored properly. *As an Employee, I want to view available products and stock so that I can assist customers easily.

Customer :

*As a Customer, I want to buy products from the shop so that I can fulfill my needs. *As a Customer, I want my basic information (name and phone) to be stored so that my purchases can be recorded.
*As a Customer, I want a quick and simple billing process so that I can save time.

Chapter 03a:
ER DIAGRAM



Chapter 03b:

SQL Queries:

```
select * from UserInfo;

 insert into UserInfo(Id, password,name, role) values('" + this.txtUserId.Text + "', '" +
this.txtPassword.Text  + "','" + this.txtName.Text + "','" + this.txtRole.Text + "');

@"update Userinfo
                        set Id = '" + this.txtUserId.Text + @"',
                        Password = " + this.txtPassword.Text + @",
                        Name = " + this.txtName.Text + @",
Role = " + this.txtRole.Text + @",                          where Id = '"
+ this.txtUserId.Text + "'; ";

delete from Userinfo where Id = '" + id + "';

select Id, name, Price,stock, ExpiryDate from Product where Id = '" + this.txtProductId.Text + "';

select * from UserInfo where Id = '" + this.txtUserId.Text + "' and Password = '" +
this.txtOldPass.Text + "';

update UserInfo set Password = '" + this.txtNewPass.Text + "' where Id = '" + this.txtUserId.Text + "';

select * from UserInfo where Id = '" + this.txtUserId.Text + "' and Password = '" +
this.txtOldPass.Text + "';

select * from Product;

SELECT Id, name, price, stock FROM Product;

UPDATE Product SET stock = " + newStock + " WHERE Id = '" + prodId + "';

SELECT MAX(BillId) FROM Bill;

SELECT [Customer Id] FROM CustomerTable;
```

```
SELECT [Customer Id] FROM CustomerTable WHERE [Customer Phone] = '" + phone + "';

INSERT INTO CustomerTable ([Customer Id], [Customer Name], [Customer Phone])
VALUES ('" + cId + "', '" + name + "', '" + phone + "')";

INSERT INTO Bill (BillId, CustomerId, CustomerName, CustomerPhone, TotalAmount) "
+"VALUES ('" + bId + "', '" + cId + "', '" + name + "', '" + phone + "', " + grandTotal + ");

select count(Id) from UserInfo where Role = 'employee';

select Id, Name,Password, Role from UserInfo where Role = 'employee';

select Id, Name,Password, Role from UserInfo where Id = '" + this.txtSearch.Text + "' and Role =
'employee';

SELECT * FROM UserInfo WHERE Id = '" + this.txtUserId.Text.ToLower() + "' AND Password = '"
+ this.txtPassword.Text.ToLower() + "';

select max(Id) from Product;

delete from Product where Id = '" + id + "';

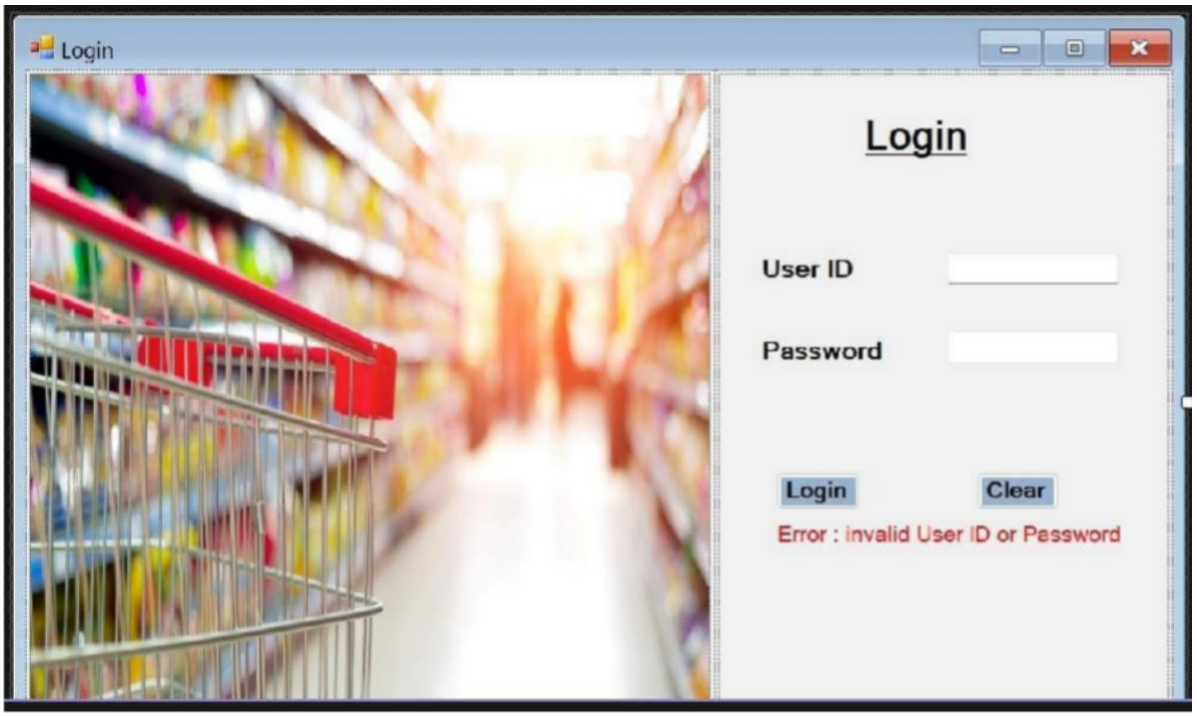
select * from Product Where Id = '" + this.txtProductId.Text + "';

insert into Product values('" + this.txtProductId.Text + "', '" + this.txtName.Text + "'," +
this.txtPrice.Text + "," + this.txtStock.Text + ",'" + this.dtpExpiryDate.Value + "');

@"update Userinfo
                        set Id = '" + this.txtProductId.Text + @"',
                        Name = '" + this.txtName.Text + @"',
                        Price = " + this.txtPrice.Text + @",
                        Stock = " + this.txtStock.Text + @",
                        Expiry Date = '" + this.dtpExpiryDate.Value + @"',
where Id = '" + this.txtProductId.Text + "'; ";

select Id, name, Price,stock, ExpiryDate from Product where Id = '" + this.txtProductId.Text + "';
```

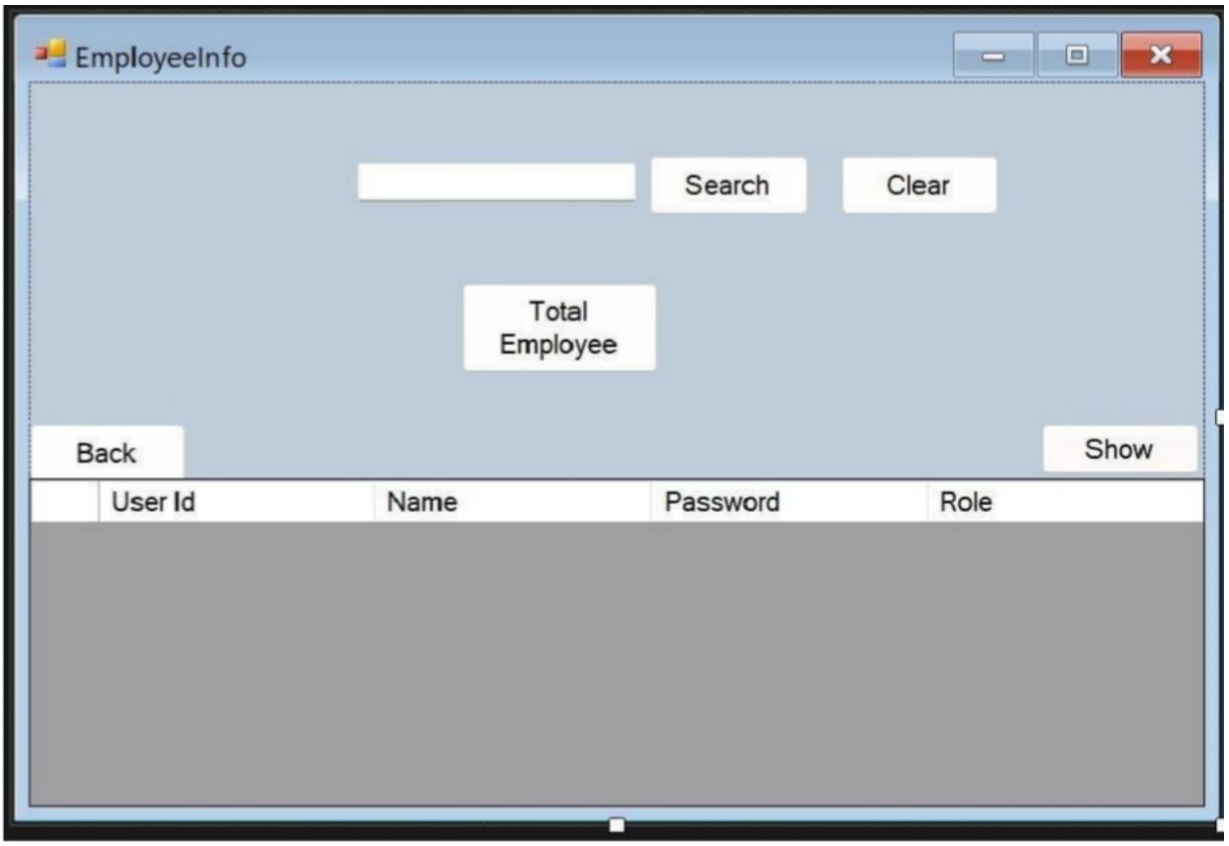
Chapter 04:


Login
Login
User ID
Password
Login
Clear
Error : invalid User ID or Password


HomeEmployee
Welcome Employee
Exit
Next

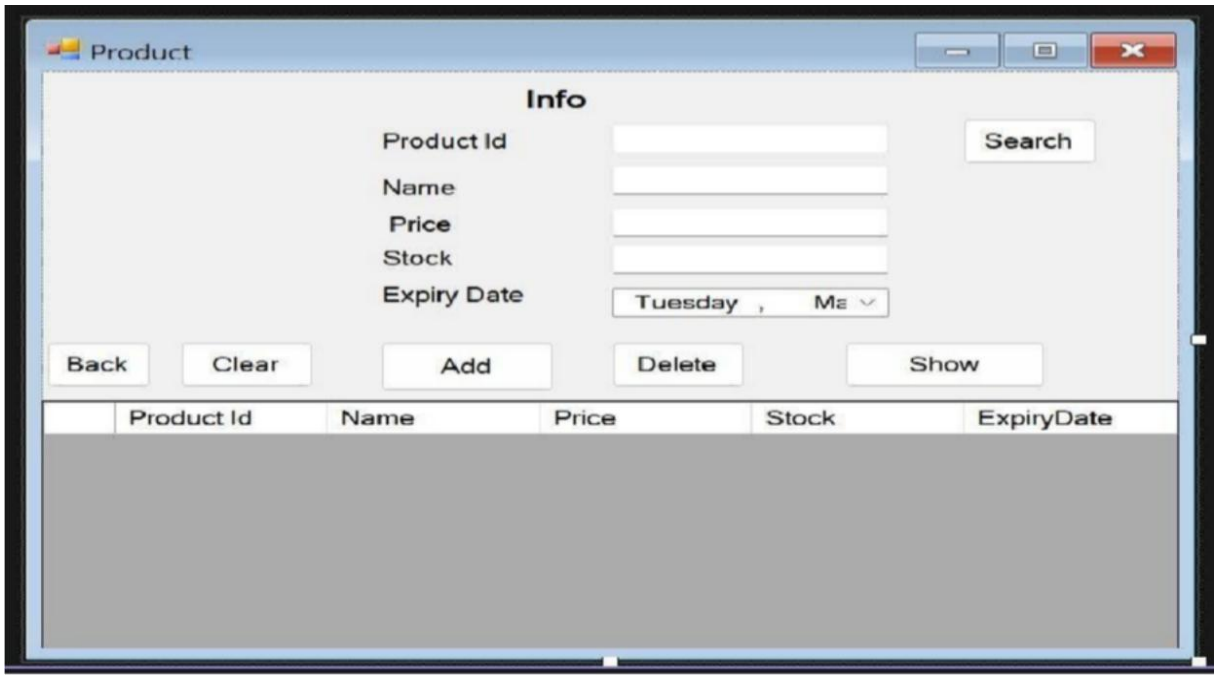

Home
Welcome Admin
Exit
Next

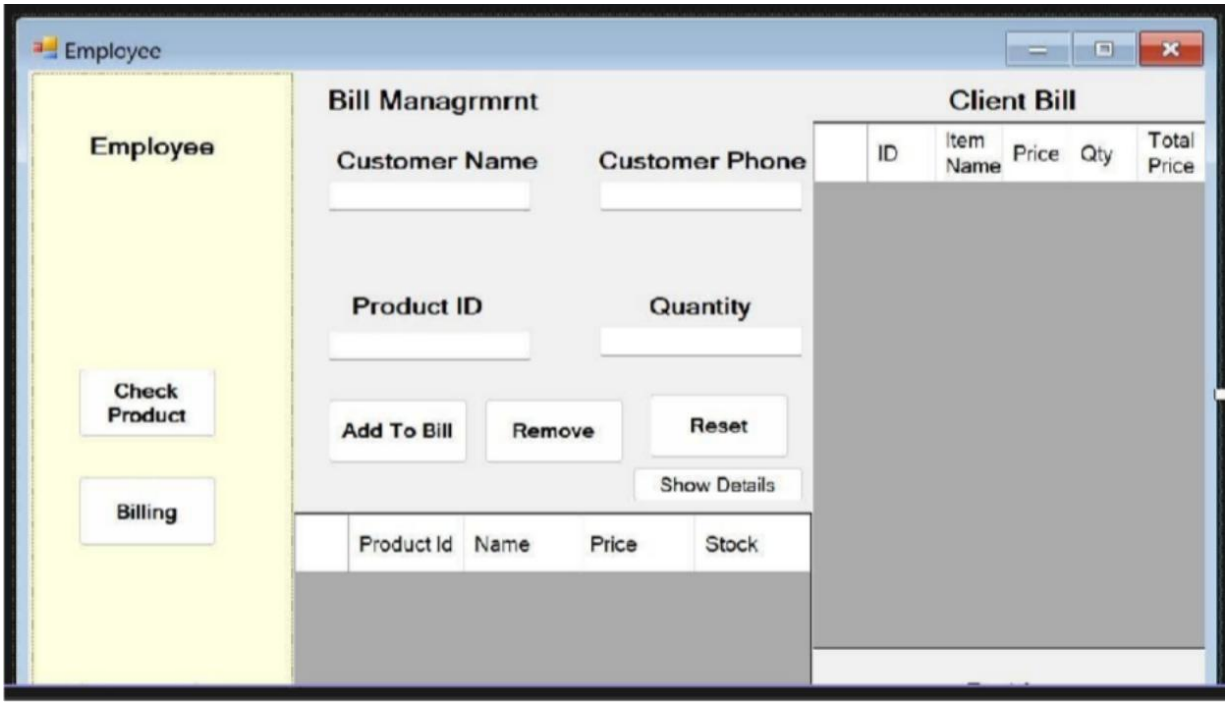

DashBoard
Dash Board
User Id
Old Pass
New Pass
Back
Update
Clear
Show All
User Id
Name
Password
Role


EmployeeInfo
Search
Clear
Total Employee
Back
Show
User Id
Name
Password
Role


Admin
Control
Add member
Dash Board
Product
Employee
Logout
Info
User Id
Password
Name
Role
Clear
Add
Delete
Show
User Id
Name
Password
Role


CheckProduct
Check Info
Product ID
Check
Back
Show Info
Product Id
Name
Price
Available
Expiry Date


Product
Info
Product Id
Search
Name
Price
Stock
Expiry Date
Tuesday ,
Back
Clear
Add
Delete
Show
Product Id
Name
Price
Stock
ExpiryDate


Employee
Employee
Bill Managrmrnt
Client Bill
Customer Name
Customer Phone
ID
Item Name
Price
Qty
Total Price
Product ID
Quantity
Check Product
Add To Bill
Remove
Reset
Billing
Show Details
Product Id
Name
Price
Stock